

Database Management Systems — Test Syllabus

Sample syllabus PDF for testing multi-subject planning.

Unit 1 — Database Fundamentals

- Database concepts and database architecture
- DBMS advantages and limitations
- Data models and schemas
- Keys and integrity constraints

Unit 2 — Relational Model

- Relational algebra
- SQL queries and joins
- Aggregate functions and grouping
- Views and nested queries

Unit 3 — Database Design

- Functional dependencies
- Normalization: 1NF, 2NF, 3NF and BCNF
- ER diagrams and mapping
- Lossless and dependency-preserving decomposition

Unit 4 — Transactions

- ACID properties
- Transaction states
- Concurrency control
- Serializability and locking

Unit 5 — Recovery and Security

- Database recovery techniques
- Log-based recovery
- Backup and restore
- Database authorization and security